



MINING DEALER BEST PRACTICE

980H Differential Brake Test

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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4.01-210311-1

1.0 Introduction

In the past, no inspection procedures were being conducted during a 980H brake test without installing the differential first. Periodically, components were delivered to customers that hadn't been tested in the workshop prior, which resulted in issues down the line. This can cost a lot of money and tarnish the customer relationship. For example, in one instance, an O-ring was damaged in a differential component, we didn't have test procedure at that time and without realizing, the component was delivered to a customer. After that, our customer couldn't run the brake and sent it back and our technicians had to spend eight hours fixing it, which cost around \$500. In addition to this financial cost, our reputation with that customer was tarnished. To avoid situations such as this, a test procedure is being rolled out in 980H brake thanks to one of our engineers.

2.0 Best Practice Description

This test procedure doesn't require the brake to be installed in the equipment. It provides an opportunity to test brake in the workshop area. The test is very easy to perform. Hydraulic oil is slowly sent through the brake with the help of a hand-operated pump.

At the same time, the pressure is checked via a pressure gauge and hydraulic oil is pumped until the pressure gauge indicates 450-500 PSI. The pressure gauge is observed for some time (approximately 1 minute), the brake test is passed if the pressure stays stable around 450-500 PSI. The test has failed if the gauge shows a dramatic decrease (1% change) from 450-500 PSI (4.5 – 5 PSI change). If the test fails, then possible problems can be investigated and fixed before returning the component to the customer. A detailed procedure is included in the sections with the addition of a video of the test in the attachments.



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3.0 Implementation Steps

Step 1:

Connect the input line of hand-operated pump with oil source. It is recommended to bleed the line beforehand to ensure that there is no air in the lines and prevent false readings.



Step 2:

Connect the output line of hand-operated pump with input line of brake.



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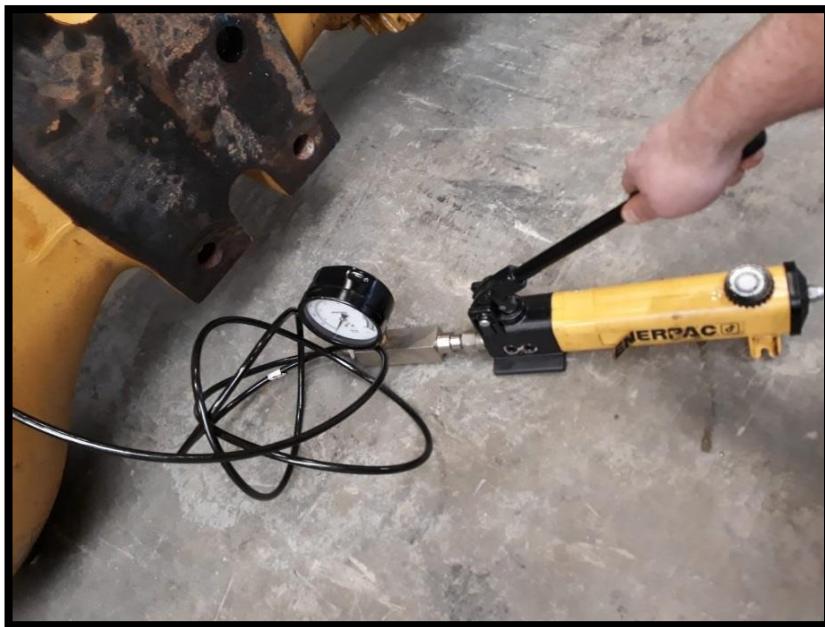
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Step 3:

Close key of the hand-operated pump and proceed to pump oil through brake.



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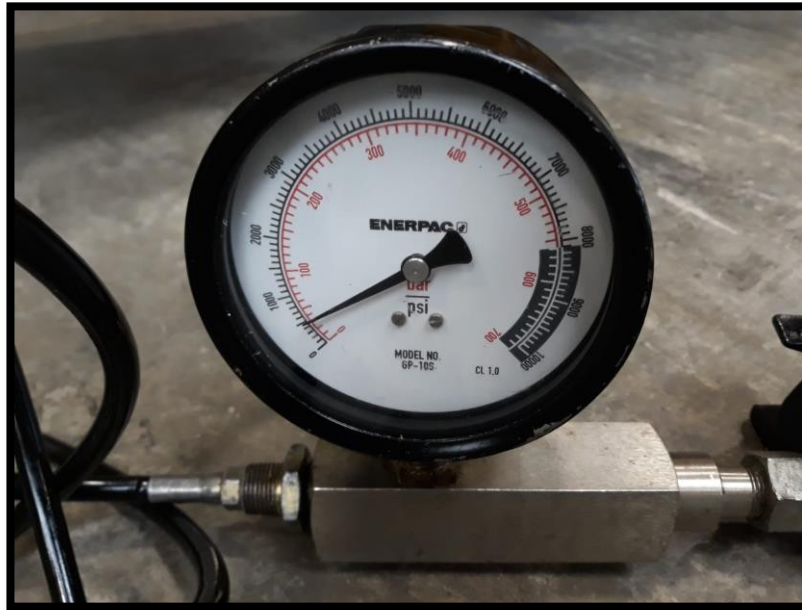
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Step 4:

Continue to supply oil until 450-500 PSI is observed. Then wait for 1 minute to observe pressure changes. The test has failed if there is more than a 1% (4.5-5 PSI) loss in pressure.



4.0 Benefits

- 1) It can save potential redo costs including cost of parts and labor.
- 2) It can save freight costs associating with rework from customer.
- 3) It increases quality of revision.
- 4) It can protect our reputation with customer by having less redo work.
- 5) It will save time by not having to install transmission to test it.

5.0 Resources Required

- Hand-operated Pump with pressure gauge

6.0 Supporting Attachments / References

- Break test. mp4 - is available - on request

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7.0 Related Best Practices

- None

8.0 Acknowledgements

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